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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Asian Sentences Is Amazingly Not Sane (ASIANS)
Selected Domain	Domain 2
Selected SDG Track (1–17)	8
Project / Solution Name	InstaQuote
Name (Leader)	Teng Min Jie
Email & Phone (Leader)	tengminjie268@gmail.com (+60182658208)
University Affiliation(s)	Asia Pacific University of Technology & Innovation
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysia's micro and small businesses are the overwhelming majority of local enterprises, yet most still find suppliers through personal contacts, WhatsApp groups and guesswork. A café owner who needs 500 eco-friendly paper bowls under RM200 often does not know whom to approach, spends hours chasing quotations, and cannot easily compare price, minimum order quantity (MOQ), delivery time and reliability. Staff may also contact suppliers without the owner's knowledge, leaving no record of who was asked, what was agreed or why.

This affects owners and small teams who lack procurement expertise, time or the budget for custom systems. The consequences are economic: overpaying, delayed stock, missed opportunities with better suppliers, and owner time lost to repetitive follow-up. Startups are hit hardest because they have no established network, which keeps them disconnected from supply chains and widens the gap with larger, better-connected firms.

Solving this supports SDG 8 (Decent Work and Economic Growth), especially targets 8.2 and 8.3, by raising SME productivity and helping small businesses grow. InstaQuote targets one recurring burden: finding, comparing and safely contacting the right supplier for a purchase.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

InstaQuote is a web application that helps Malaysian SMEs discover, compare and contact suppliers through a human-controlled workflow. Staff describe a need in plain language, for example "500 eco-friendly paper bowls in Selangor under RM200". The system turns this into a structured request, searches a supplier directory, filters by product fit, budget, MOQ, location and delivery time, and ranks the Top 3 suppliers by price, delivery speed, reliability and

compatibility. Every recommendation, and every rejection, comes with a plain-language explanation.

AI is used only where it adds value. A GenAI API interprets natural language requests into form fields and drafts an editable introduction or RFQ in English or Bahasa Melayu. Matching and ranking use transparent rules and weighted scoring, so results are accurate, consistent and explainable rather than coming from a black-box chatbot.

People stay in control. A role switcher simulates Staff and Owner permissions. Staff submit the draft for approval, and only the Owner can approve or reject it. On approval, the system generates a prefilled WhatsApp link so the Owner contacts the supplier manually. Every request, decision and outreach status is recorded in an audit log.

We intend to leverage [GenAI APIs for language understanding and RFQ drafting, a database for supplier and audit data, and cloud hosting for the live demo]. All data is synthetic. As a reusable module, InstaQuote can plug into the Orbyte business hub and serve many SMEs.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Our conceptual tech stack uses Python and Streamlit for the Orbyte-styled SaaS frontend, Pandas for supplier evaluation, a PostgreSQL database (psycopg2) with a resilient local fallback (suppliers.csv), and a zero-dependency urllib REST client calling cloud GenAI APIs (Gemini, Groq, and OpenRouter) alongside wa.me deep links for WhatsApp RFQ dispatch. In our system block diagram, the user interacts with the Streamlit web app hosted on Replit, which pulls low-stock triggers from the Orbyte hub, queries the database/CSV to rank 36 verified Malaysian suppliers, sends a token-capped prompt to the GenAI API to draft bilingual RFQs, enforces Role-Based Access Control (Staff draft vs. Owner approval), and logs audit events while generating the final WhatsApp link. For official SDC sponsor-provided platforms, our project exclusively utilized Replit (Replit Core) as our cloud development environment and live deployment host.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We applied the five-stage design-thinking methodology to ensure Orbyte InstaQuote is practical and deployment-ready:

1. **Empathize:** We examined how Malaysian F&B and retail SMEs lose hours manually messaging vendors on WhatsApp during stockouts.
2. **Identify:** We pinpointed two core bottlenecks: unstandardized vendor comparison (MOQ, lead time, budget) and unauthorized staff purchasing without owner oversight.
3. **Ideate:** We designed an automated workflow combining explainable 100-point supplier scoring, bilingual GenAI RFQ drafting, and role-based human-in-the-loop approval.

4. **Prototype:** We built a cohesive procurement module integrated into the Orbyte hub UI on Replit, featuring a dual-layer data architecture (PostgreSQL with automatic CSV/session fallback) and zero-persistence API key handling.
5. **Test:** We validated multi-tenant switching, RBAC permissions, and token-safe AI rate limits. Hosted live on Replit Deployments, the application is stable, scalable, and ready for Phase 2.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 8, Target 8.3: "Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalization and growth of micro-, small- and medium-sized enterprises, including through access to financial services." Secondary: Target 8.2, higher economic productivity through technological upgrading and innovation.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	MSMEs make up about 97.4% of Malaysian business establishments (1,173,601 of 1,204,987 in 2022; 78.7% are micro). They contributed 39.7% of GDP and 48.7% of employment in 2025. Despite this weight, supplier sourcing is still manual (WhatsApp, personal contacts, calls). Team estimate: one sourcing task takes about 3 hours of active staff time (finding suppliers, comparing quotes, drafting messages) and about 30% of purchases are made without comparing at least three suppliers.
Core Measurable KPI *The specific unit you will track*	Active staff time (minutes) from request to an owner-approved, ready-to-send supplier message, per sourcing task. Supporting KPIs: percentage of requests that compare

Metric Category	Your Response
e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.	3 or more suppliers, and percentage of outreach actions with an audit record and Owner approval.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	At least a 70% reduction in active sourcing time (from about 180 minutes to 55 minutes or less) in prototype testing with [5] test users on [5] synthetic scenarios. Also 100% of recommendations explained, 100% of outreach approved by the Owner and logged, and at least 90% of requests compared across 3 suppliers.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ✓ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ✓ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ✓ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]



[Date]

24/10/2026